

Complications of BCG Vaccination in Neoplastic Disease

C. WILLIAM AUNGST, M.D., F.A.C.P., JOSEPH E. SOKAL, M.D., F.A.C.P., and

BLAIR V. JAGER, M.D., Buffalo, New York

More than 300 patients with neoplastic disease (most with lymphoma or leukemia) were given one or more BCG vaccinations. In the great majority of cases, these were well tolerated. The complications we observed were of three types: (A) persistent BCG infection that could disseminate widely, (B) activation of old, dormant acid-fast infection, and (C) hypersensitivity reactions. The latter were related to the frequency of administration and dose of organisms. It is important to recognize that BCG organisms may not be completely destroyed by patients with impaired immunologic defenses, and that their inoculation may create a source of disseminated acid-fast infection at a later time. In our limited experience, treatment with isoniazid for 2 to 3 months has resulted in cure of BCG infection. Hypersensitivity reactions may constitute a more serious problem than such infections.

INOCULATION of living BCG organisms, in doses recommended for the prevention of tuberculosis, is relatively free of side-effects, and millions of BCG vaccinations have been done without incident. Most of the severe reactions have occurred in infants or young children, or when mistakes have been made in the concentration of organisms in the vaccination material (1-3).

During the past few years, a new use of BCG has received considerable attention. As a result of animal experiments in many laboratories, which have shown effectiveness of these organisms in increasing host resistance to a variety of tumors, BCG is now being investigated as a therapeutic agent for human cancer. The use of BCG for immunotherapy of cancer differs in several respects from its use to immunize population groups against tuberculosis: (A) Repeated vaccinations are given, sometimes at intervals as short as 5 days. (B) Substantially larger doses of organisms may be administered. (C) Organisms are often injected directly into tumor masses, rather than being inoculated superficially into the skin. Because many of the injected tumors are quite vascular, this may deliver organisms into the circulation; BCG bacteremia has been shown after such injections (4). (D) Unlike normal individuals receiving BCG vaccination as prophylaxis against tuberculosis, many of the patients with neoplastic disease have depressed immunologic reactivity

and, in addition, may be receiving simultaneous antitumor therapy with immunosuppressive agents.

In the course of almost 2 decades of experimental studies of the effects of BCG vaccination among patients with malignant neoplasms, we have found that inoculation of these organisms is generally well tolerated. However, we have encountered a number of side-effects and complications.

Clinical Materials and Methods

We have administered BCG vaccine to more than 300 patients with neoplastic disease. More than half of them had malignant lymphoma in various stages. Most of the remainder had chronic lymphocytic or myelocytic leukemia. Other diagnoses represented in this series include osteogenic sarcoma, malignant melanoma, and a variety of carcinomas. Approximately 100 patients have received repeated vaccinations with BCG alone, or with BCG-cell mixtures, at intervals ranging from 1 week to 1 year.

With a few exceptions, BCG was not administered to patients receiving prednisone, radiation therapy, or multiple-agent chemotherapy. The great majority of patients were vaccinated with organisms of the Tice strain (Research Foundation, Chicago, Illinois) administered by intradermal injection in volumes ranging from 0.1 to 0.6 ml. Doses of organisms administered ranged from 0.004 to 2.4 mg (moist weight), containing 4×10^4 to 25×10^6 colony-forming units. These dosages reflect material actually injected into the patient. (With other techniques of vaccination, such as scarification and multiple puncture techniques, much larger numbers of organisms are applied to the skin and listed in descriptions of "dose." However, only fractions of the organisms in such doses are actually inoculated into the patient.)

Results

Although there were some differences between local reactions to injection of BCG-cell mixtures and of BCG alone (5), the principal side-effects and complications we observed were attributable to the BCG content of our vaccines. Accordingly, both types of vaccination are considered together.

The usual reaction to the first intradermal injection of BCG consists of formation of an erythematous papule that breaks down, forming a shallow ulcer. This may drain for some days. Later, an adherent scab forms and there is gradual healing over a period of 3 to 6 weeks. The vaccination leaves a permanent scar that closely approximates the area into which organisms were injected. Occasionally, enlargement of the lymph nodes draining the vaccination site, with or without tenderness, may be noted as the reaction is evolving. There is virtually no morbidity associated with these events.

► From the Departments of Medicine B and Pathology, Roswell Park Memorial Institute, New York State Department of Health, Buffalo, New York.

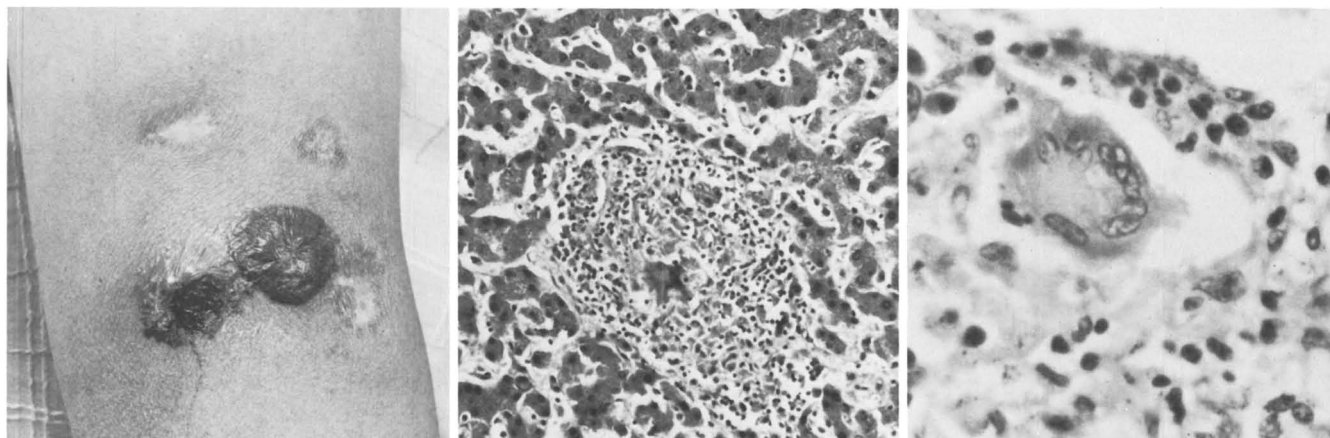


Figure 1. Left. Upper arm of a highly sensitized patient, 8 days after BCG vaccination. Cellulitis, which involved a large area of the arm, has subsided. There is hemorrhagic necrosis at the recent vaccination site. Middle. Miliary granuloma in liver, Patient 6 (hematoxylin and eosin stain; original magnification, $\times 150$). Right. Miliary granuloma in lung, Patient 6 (magnification, $\times 500$). Acid-fast stains of these specimens were negative, but BCG organisms were recovered on culture.

COMPLICATIONS AMONG IMMUNOCOMPETENT PATIENTS

When immunocompetent patients received repeated vaccinations, the local reactions became more severe. Cellulitis, extending several centimetres around the vaccination site, developed within 24 hours and lasted for several days. Vaccination ulcers were deeper, drainage was more profuse, and healing was slower. About half of the immunocompetent patients who received multiple vaccinations developed systemic reactions. These were usually not seen until after at least three to four vaccinations. Typically, they started 6 to 10 hours after a vaccination, with relatively sudden onset of malaise, chilliness or shaking chills, fever of 38 to 39 °C, and disabling symptoms for 5 to 7 days. In one such case, the patient developed a large hemorrhagic bleb at the vaccination site and in a neighboring previously healed site, in addition to a prolonged febrile course. Figure 1 (*left*) shows this patient's arm 8 days after vaccination.

Other hypersensitivity reactions to BCG are shown in Table 1. Two patients (Patients 1 and 2) with strong delayed hypersensitivity responses developed erythema nodosum after their second and tenth vaccinations, respectively. This complication was not encountered in subsequent vaccinations, after threefold to sixfold reductions in BCG doses. One of these (Patient 2) also developed pancytopenia (while off antileukemic therapy) with purpura. A third patient (Patient 3) had activation of a dormant acid-fast infection with an atypical *Mycobacterium* after a prolonged febrile hypersensitivity reaction.

REACTIONS AMONG IMMUNODEPRESSED PATIENTS

Patients with severe depression of cellular immune responses, due either to their neoplastic disease or to treatment with immunosuppressive agents, do not respond normally to BCG vaccination. We have seen several patients of this type, vaccinated by ourselves or others, who have shown no local reaction to BCG vaccination at all. Other patients have had abortive reactions with formation of a small papule at the injection site, which regressed without ulceration or scarring.

It is among immunodepressed patients that persistent and possible serious acid-fast infection may occur after BCG vaccination. Two such cases are summarized in Table 1. In one of them (Patient 5), definitive identification of the acid-fast organism was not accomplished, but the circumstantial evidence strongly implicated BCG as the cause of the "tuberculous" infection. In the other case (Patient 6), acid-fast organisms were cultured from autopsy specimens of liver and lung (Figure 1, *middle and right*), and these were identified as BCG*. It is noteworthy that in both cases, the local reaction at the BCG injection site evolved normally, with complete healing. Evidence of active acid-fast infection appeared some months later.

Discussion

At present, treatment of neoplastic diseases with BCG remains experimental and should be restricted to controlled clinical trials. It must be emphasized that the troublesome complications we have encountered involved only a small fraction of the total number of patients injected with BCG. These complications fell into three categories: (A) establishment of persistent acid-fast infection, which could disseminate widely and cause serious pathology; (B) activation of previous, dormant acid-fast infection; and (C) hypersensitivity reactions.

Although induction of disseminated infection, which could lead to death from "miliary tuberculosis," might appear to be the most serious hazard of administration of these living bacteria, we believe that this complication can be prevented easily if appropriate precautions are taken. The BCG organism is highly sensitive to the drugs commonly used for treatment of tuberculosis (isoniazid, streptomycin, ethambutol, rifampin), as well as to drugs with lesser activity, such as kanamycin. In our limited experience, 2 to 4 months of therapy with a well-tolerated drug, isoniazid, has been sufficient to eradicate BCG infection. The patients most susceptible to BCG infection are those

* Bacteriologic studies of these specimens were done by Mr. Nat Minsley, supervisor microbiologist, Veterans Administration Hospital, Buffalo, New York.

Table 1. Examples of Unusual Reactions to BCG Vaccination

Patient	Diagnosis	Summary of Clinical Data
Hypersensitivity reactions		
1	Chronic myelocytic leukemia	History of tuberculosis exposure; positive first-strength PPD; marked local reaction to first vaccination; ill for 2 days after second vaccination. Later, swelling and pruritic rash of right arm and, then, erythema nodosum of legs that subsided in 1 week without treatment.
2	Chronic myelocytic leukemia	Strong delayed skin-test responses; febrile reaction lasting 24 hours after tenth vaccination. One-week later, chills and fever, profuse drainage from vaccination site, and flare-up of old PPD skin-test site. One-month illness with prostration, 11.34-kg [25-lb] weight loss, and pancytopenia with purpura. Erythema nodosum 4 weeks after vaccination. Gradual recovery without specific therapy.
Activation of dormant acid-fast infection		
3	Chronic myelocytic leukemia	Strong delayed skin-test responses; fever for 10 days after his 24th vaccination. Ten days later he developed a profusely draining vaccination ulcer, tender fluctuant axillary adenopathy, and fever of 38.6 °C. Therapy with isoniazid failed. Incision and drainage done; cultures showed <i>Mycobacterium terrae</i> , resistant to isoniazid. Patient recovered after therapy with ethambutol.
Reactions in immunodepressed patients		
4	Cancer of adrenal	Started on oral cyclophosphamide the same day as BCG vaccination. No evidence of any reaction at the vaccination site until 3 weeks after the 4-week course of this immunosuppressive drug was completed; then had a typical local reaction, with formation of a shallow ulcer.
5	Hodgkin's disease	Stage III B disease previously treated with radiation and nitrogen mustard. Treated with prednisone for hemolytic anemia. May 1961; BCG vaccination, PPD transiently positive 3 weeks later; radiation therapy December 1961; Mental deterioration and intermittent fever March 1962 until death in July 1962. Break-down of previously healed BCG vaccination site. Autopsy showed multifocal leukoencephalopathy and Hodgkin's disease; acid-fast bacilli were shown in the ulcer at the BCG injection site and the draining lymph nodes; also, ipsilateral "tuberculosis" empyema and a paraspinal cold abscess were found.
6	Melanoma	Melanoma diagnosed in 1967; progressive disease, 1970. Therapy with imidazole and actinomycin D in 1971. PPD skin test positive when vaccinations with BCG and cultured melanoma cells were given in January and February 1972. Developed pulmonary and osseous metastases, and fever. Brief course of hydrocortisone. Died May 1972 with extensive tumor and "miliary tuberculosis" involving the liver, lungs, spleen, and bone marrow (Figure 1, middle and right). Cultures of liver and lung grew BCG organisms.

with impaired delayed hypersensitivity responses. However, complete anergy is not a prerequisite for such infection.

To avoid potentially serious infections, the following points must be kept in mind: (A) Absence of a local reaction to vaccination does *not* mean that viable organisms were not introduced. (B) Complete healing of a vaccination site does not mean that BCG organisms in that location have been permanently eradicated. (C) The manifestations of disseminated acid-fast infection may be extremely difficult to distinguish from those of actively progressing neoplastic disease.

It is now our policy to treat patients promptly with isoniazid, with or without streptomycin, for any clinical findings consistent with acid-fast infection. We also administer isoniazid prophylactically to previously vaccinated patients who develop advancing neoplastic disease that may impair their immunologic defenses, and, especially, if they are treated with immunosuppressive agents.

Our principal concern is with hypersensitivity reactions after repeated administration of BCG to immunocompetent patients. The most common of these, a prompt febrile reaction lasting 1 to 2 days, has not posed any significant problems for us and is acceptable to our patients. However, reactions of the type encountered in Patient 2 (Table 1) are very disturbing. This busy physician was disabled for a month by an unpredictable immunologic disease that produced multiple systemic abnormalities, including substantial leukopenia and severe thrombocytopenia. Had this reaction been somewhat more severe, it might have threatened his life.

The risk of various complications of BCG vaccination

may differ according to the technique of inoculation. Scarification and multiple puncture techniques appear to be associated with less morbidity than intradermal injection. Intratumoral injection is certainly the most hazardous, because it is likely to result in bacteremia (4) and, thus, could produce either disseminated infections among immunodepressed patients or serious hypersensitivity reactions among previously sensitized immunocompetent patients. Recently, two fatal reactions following intralesional injection of BCG in highly sensitized patients were reported*†. The course of both was similar, with chills and cyanosis early, and then pyrexia with hypotension at 6 hours. These patients died within 3 days, with pancytopenia and evidence of disseminated intravascular coagulopathy. Other hazards of intralesional injection include a significant incidence of hepatic dysfunction and the development of noncaseating granulomas of the liver (6).

* SPITLER L, WYBRAN J, LIEBERMAN R, et al: *Results of Intraleisional BCG Therapy in Malignant Melanoma: Clinical and Immunologic Evaluations and Complications*. Presented at the BCG Conference, Chicago, Illinois, 5 September 1973.

† MCKHANN C, GUNNARSON A: *A Fatality Secondary to Hyperimmunity to BCG*. Presented at the BCG Conference, Chicago, Illinois, 5 September 1973.

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► Requests for reprints should be addressed to C. William Aungst, M.D., Associate Director for Clinical Affairs, Roswell Park Memorial Institute, 666 Elm Street, Buffalo, NY 14203.

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"Dominion of the . . . Resistive Mind"

Throughout the enormously long period during which modern man has been established on the earth human society has been left to the uncontrolled contention of constructive and destructive forces, and in the long run the destructive have always proved the stronger. Whether the general level of consciousness will reach the height necessary to give a decisive predominance to constructive tendencies, and whether such a development will occur in time to save Western civilization from the fate of its predecessors, are open questions. The small segment of the social process of which we have direct knowledge in the events of the day has no very encouraging appearance. Segregation has reasserted itself effectively; the dominion of the stable and resistive mind is as firmly established as ever, and no less dull and dangerous; while it is plain how far, in the atmosphere of relaxation and fatigue, the social inspiration of the common man has sunk from the high constancy of spirit by which throughout the long pilgrimage of war so many weary feet have been upborne, so many dry lips refreshed.

W. TROTTER
Instincts of the Herd in Peace and War
 London, T. Fisher Unwin Ltd., 1916, p. 259